

Ishaan Virmani

DATA & AI ENGINEER

✉ ivirmani@gmail.com

📍 Haryana, India.

🌐 linkedin.com/in/ishaan-virmani

☎ +61-422115681 / +91-9996162945

🌐 ivirmani.github.io/ishaan_portfolio

🐙 github.com/ivirmani

Seasoned Data and AI engineer with 3+ years' industry experience across fintech, aerospace, and independent consulting, building ETL pipelines, optimizing SQL for large-scale reporting, and maintaining data quality through automated validation and testing. Skilled at translating stakeholder requirements into technical solutions and working with large language models (LLMs), including Claude and GPT, Azure Foundry, and agentic AI workflows. Comfortable operating in regulated environments where accuracy and traceability matter.

WORK EXPERIENCE

PRIVATE TRUST | Feb 2024 - Present

Data Integration Specialist | Brisbane (Remote)

- Built ETL pipelines processing 10M+ records, streamlining reporting and analytics workflows.
- Surfaced business-critical spending trend shifts and anomalies performing exploratory data analysis (EDA) on 100K+ client transactions.
- Created AI agents using Azure Foundry to support reporting and analytics workflows.
- Configured agent instructions and data connections for contextual business responses.

MINTOAK INNOVATIONS PVT. LTD. (B2B SaaS - Fintech) | Dec 2025 - Mar 2026

Data Science Analyst | Melbourne - Remote

- Enabled banking clients to reduce manual reconciliation effort by ~4 hours/week.
- Improved system usability by 35% as measured by internal KPIs by developing in-house automated recurring analytics reports for banking clients (leveraging Apache Superset and Power BI).
- Optimised complex analytics queries, KPI logic and data transformations, reducing reporting inconsistencies by 15%+.
- Expertly translated stakeholder requirements (business, customers) into technical analytics solutions, cutting feedback-to-release cycles by 20%.

UNIVERSITY RESEARCH PROJECT - The Universal Book | Nov 2024 - Feb 2025

Machine Learning Intern (Guided by Prof. Levin Kuhlmann) | Melbourne

- Collaborated with Prof. Levin Kuhlmann (Monash University) on a faculty-led research project applying NLP to detect hate speech and plagiarism in user-submitted content.
- Designed and built NLP pipelines covering preprocessing, feature extraction, and model training workflows for automated content screening.
- Applied a Hugging Face hate speech detection model to automatically screen author-submitted book content prior to publishing on a web platform, flagging harmful language pre-publication.
- Fine-tuned Toxic-BERT (Hugging Face) for anomalous speech detection, cutting inference time by 20% (from ~250ms to ~200ms per prediction) through batch size tuning and input sequence truncation.

BOEING | Aug 2022 - Jan 2024

Associate Software Engineer | Bengaluru

- Contributed to 3 major teams within Boeing's Training Solutions domain (BCA & BGS), collaborating with 40+ developers to deliver pilot training and assistance software used across commercial aircraft training programs.
- Co-developed REST APIs and data processing pipelines with senior engineers, owning key architectural decisions for ingesting and transforming real-time flight training data, enabling reliable data consumption across multiple downstream training systems.

EDUCATION

MONASH UNIVERSITY | Feb 2024 - June 2026

Master of Data Science | Melbourne | Ranked #31 in QS Rankings 2027, #5 in Australia

Relevant Coursework: Big Data, Data Restoration, Data Cleaning, Data Engineering, Spatial Databases, Data Visualisation, Data Wrangling, Data Processing, Python, Machine Learning, Data Science

Extracurriculars: Monash Data & AI Society (Member) | Monash Student Association & Monash Graduate Association (Event Volunteer)

Recognition: **MONASH STUDENT CHOICE AWARD 2026 WINNER FOR INDUSTRY EXPERIENCE**

M.S. RAMAIAH INSTITUTE OF TECHNOLOGY (MSRIT) | Aug 2018 - Jul 2022

B.E in Information Science and Engineering | Bengaluru, India

Relevant Coursework: DSA, Operations research, Cloud, Computer Science, Computing, Data Science, Machine Learning

Extracurriculars: CodeRIT (Leader) | TEDxMSRIT (Volunteer) | Developer Student Club (Member) | NumeraRIT (Member) | iClick (Member)

PERSONAL PROJECTS, PORTFOLIO & VOLUNTEERING

Relo'Oz - Suburb Relocation Platform for Vulnerable Groups (Monash University) - STUDENT CHOICE AWARD WINNER - <https://www.relo-oz.app/>

Developed a suburb relocation platform by integrating 15+ open government datasets (Census, SEIFA, rental, crime, transport) in Python (pandas, geopandas), unified on a common suburb key via spatial joins. Designed a recommendation engine matching normalised suburb feature vectors to user preferences via Euclidean distance and built an ML rent-forecasting model (Lasso; MAE ~\$28/week, R^2 0.96) projecting weekly rents to 2030.

The December Effect - Retail Seasonality Analysis (Kaggle) - <https://www.kaggle.com/code/ishaanvirmani99/the-december-effect>

Analysed 40+ years of ABS retail data, isolating the seasonal effect from raw vs. seasonally-adjusted figures, to quantify December's sales spike and identify which retail categories are most exposed to it.

YouTube Analytics Platform for Streamers (In-progress) - <https://streamdash.dev/>

Designed and developed a web-based analytics dashboard providing actionable guidance for YouTube streamers on optimal content topics and streaming schedules to maximise audience discovery, engagement, and retention.

Analytics Dashboard for Health Sector and AQI Data of New York (Monash University)

Developed an R Shiny application to explore hospitalisation trends across New York and their correlation with air pollution using PCA, treemaps, and combined visualisations, turning complex data into actionable outcomes for policymakers and health officials.

SKILLS

Data & Integration: Apache Spark; Kafka; PySpark; ETL/ELT Pipelines; API Integrations (REST, webhooks); PostgreSQL; Data Warehousing; Cloud Data Integration

Machine Learning & AI: Agentic AI; Machine Learning; Azure AI Foundry

Cloud, BI & Data Platforms: Power BI; Apache Superset; Tableau; Databricks; AWS; Microsoft Azure; Excel

Programming & Analytics: Python (pandas, NumPy, scikit-learn); SQL; R; Statistical Analysis; Forecasting; Regression modelling

CERTIFICATIONS

AI-901 Microsoft Azure AI Fundamentals (Ongoing); Generative AI with Large Language Models (AWS and DeepLearning.AI); Microsoft Azure AZ-900 certification; Machine Learning (Stanford University - Coursera) by Andrew Ng; Databricks Fundamentals

INTERESTS

Guitar, Soccer, Swimming, Photography